Barcode Recognition

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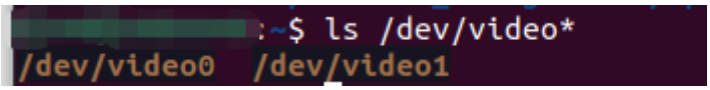
Hardware Connection

Plug the USB camera into the motherboard's USB port. Open the system terminal and enter the following command to check the camera's device number. If /dev/video0 appears, it's working correctly.

```
ls /dev/video*
```



Install Driver Libraries

```
sudo apt install libzbar-dev
sudo pip3 install pyzbar
sudo pip3 install python-barcode
```

Example Code

```
import cv2 as cv
from pyzbar.pyzbar import decode
import time
import numpy as np
from PIL import Image, ImageDraw, ImageFont


def decodeDisplay(image):
    gray = cv.cvtColor(image, cv.COLOR_BGR2GRAY)
    barcodes = decode(gray)
    for barcode in barcodes:
        # 提取条形码的边界框的位置
        # Extract the position of the boundary box of the barcode
        (x, y, w, h) = barcode.rect
        cv.rectangle(image, (x, y), (x + w, y + h), (225, 0, 0), 5)
        encoding = 'UTF-8'
        # 画出来，就需要先将它转换成字符串
        # to draw it, you need to convert it to a string
        barcodeData = barcode.data.decode(encoding)
        barcodeType = barcode.type
        # 绘出图像上数据和类型
```

```
        # Draw the data and type on the image
        pilimg = Image.fromarray(image)
        # 创建画笔
        # create brush
        draw = ImageDraw.Draw(pilimg)  # 图片上打印  Print on picture
        # # 参数1：打印坐标，参数2：文本，参数3：字体颜色
        # Parameter 1: print coordinates, parameter 2: text, parameter 3: font
color
        draw.text((x, y - 25), str(barcode.data, encoding), fill=(255, 0, 0))
        # # PIL图片转cv2 图片
        # PIL picture to CV2 picture
        image = cv.cvtColor(np.array(pilimg), cv.COLOR_RGB2BGR)
        # 向终端打印条形码数据和条形码类型
        # Print barcode data and barcode type to terminal
        print("[INFO] Found {} barcode: {}".format(barcodeType, barcodeData))
    return image


if __name__ == '__main__':
    capture = cv.VideoCapture(0)
    cv_edition = cv.__version__
    if cv_edition[0] == '3': capture.set(cv.CAP_PROP_FOURCC,
cv.VideoWriter_fourcc(*'XVID'))
    else: capture.set(cv.CAP_PROP_FOURCC, cv.VideoWriter.fourcc('M', 'J', 'P',
'G'))
    capture.set(cv.CAP_PROP_FRAME_WIDTH, 640)
    capture.set(cv.CAP_PROP_FRAME_HEIGHT, 480)
    print ("capture get FPS : ",capture.get(cv.CAP_PROP_FPS))
    while capture.isOpened():
        start = time.time()
        ret, frame = capture.read()
        action = cv.waitKey(10) & 0xFF
        frame = decodeDisplay(frame)
        end = time.time()
        fps = 1 / (end - start)
        text = "FPS : " + str(int(fps))
        cv.putText(frame, text, (30, 30), cv.FONT_HERSHEY_SIMPLEX, 0.6, (100,
200, 200), 1)
        cv.imshow('frame', frame)
        if action == ord('q') or action == 113: break
    capture.release()
    cv.destroyAllWindows()
```

Running the Program

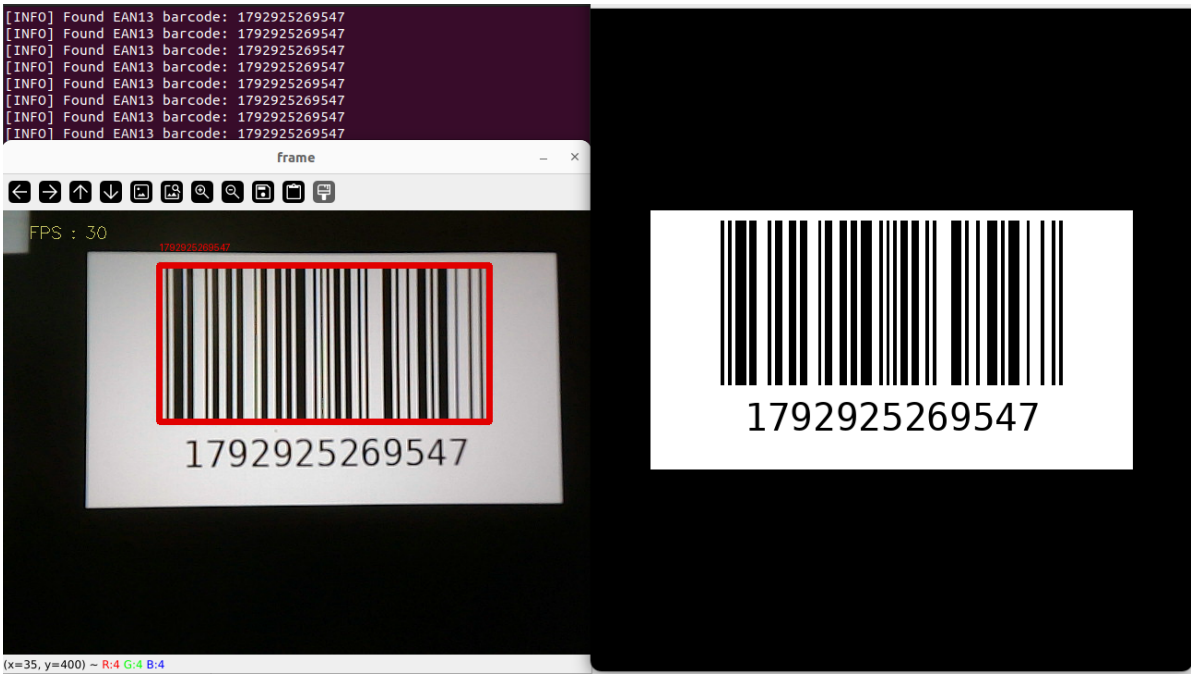
Open your desktop terminal and run the program:

```
cd ~/content_recognition/barcode
python3 barcode_parsing_usb.py
```

Note: Because the program requires a display screen, do not run it in a remote SSH terminal. Please use the native desktop or VNC remote desktop to run the program.

Program Results

After the program runs, it automatically opens the USB camera. Point the camera at the barcode, and it will outline the barcode and print the barcode information in the terminal.



Barcode Example Image:

Optional Operation: Generate Barcode

Open the terminal and run the following command to regenerate the barcode in the code directory.

```
cd ~/content_recognition/barcode
python3 barcode_create.py
```

content_recognition / barcode / code
ean13_1.
png
ean13_2.
png
ean13_3.
png